

- 4 (a) Explain what is meant by a *field of force*.

A region (of space) where a force is experienced by a body.

[1]

- (b) State **one** similarity and **one** difference between the fields produced by an isolated point charge and by an isolated point mass.

similarity: Both fields are pointing in radial direction or the strengths of both fields are inversely proportional to the square of distance from the source.

difference: For charge, the field can be away or toward the charge, but for mass, the field can only be toward the mass. Do not accept repulsive/attractive field/force/effects of field.

[2]

- (c) An isolated solid metal sphere A of radius R has charge $+Q$, as illustrated in Fig. 4.1.

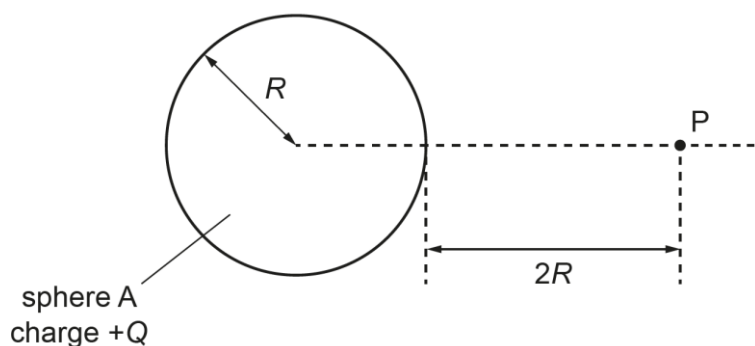


Fig. 4.1

A point P is distance $2R$ from the surface of the sphere.

Determine an expression that includes the terms R and Q for the electric field strength E at point P.

$$E = \frac{Q}{4\pi\epsilon_0 x^2} = \frac{Q}{4\pi\epsilon_0 (3R)^2} = \frac{Q}{36\pi\epsilon_0 R^2} \quad [A1]$$

$$E = \dots\dots\dots [1]$$

- (d) A second identical solid metal sphere B is now placed near sphere A. The centres of the spheres are separated by a distance $6R$, as shown in Fig. 4.2.

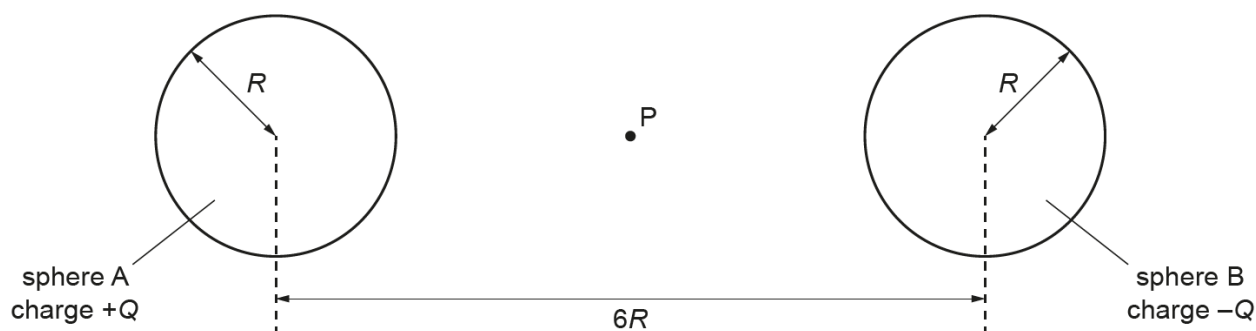


Fig. 4.2

Point P lies midway between spheres A and B.

Sphere B has charge $-Q$.

Explain why

- (i) the magnitude of the electric field strength at P is given by the sum of the magnitudes of the field strengths due to each sphere.

The fields due to each sphere are in the same direction.

.....[1]

- (ii) the electric field strength at point P due to the charged metal spheres is not, in practice, equal to $2E$, where E is the electric field strength determined in (c).

The charges on the spheres attract each other / charge distribution on each sphere

distorted by the other sphere [B1]

The charges on the spheres does not appear to act from the centres of the spheres /

distance between charges to P is not equal to $3R$ [B1]

.....[2]

[Total: 7]

5 A battery of electromotive force (e.m.f.) 9.0 V and internal resistance $2.0\ \Omega$ is connected to a